Where to Place Dropout with Batch Normalization in Residual Networks (2023)

1.

Why can naively combining Dropout with Batch Normalization in residual networks hurt training or generalization, and what interaction between the two techniques causes this?

Dropout creates a mismatch between training and inference because it increases feature variance during training but is removed during inference. Although the mean remains consistent, the variance does not. When Dropout is placed before BatchNorm, BN learns its statistics from the higher training-time variance and then receives a different variance during inference. This causes BN to normalize the features inconsistently between training and testing. This variance inconsistency can reduce training or generalization performance.

2.

Where do the authors recommend placing Dropout relative to BatchNorm and residual connections, and what theoretical or empirical reasoning supports that placement?

The paper recommends placing one Dropout after the last BatchNorm and before the last weight layer in the residual branch, such as P5 or P6. This prevents Dropout from directly affecting the input statistics of a following BN layer. The residual connection then helps reduce this variance inconsistency by combining the residual and skip branches before the next BatchNorm. Residual connections can mitigate this mismatch better than non-residual blocks. In the prediction head, they additionally recommend placing Dropout after BN but before Global Average Pooling (GAP) because it produces a more stable output than placing Dropout after GAP.

3.

How does the paper analyze the effect of Dropout on the mean and variance statistics used by BatchNorm during training versus inference?

During training, Dropout randomly removes features and scales the remaining ones by (1/p). This keeps the expected mean approximately consistent with inference. However, the variance becomes larger during training than during inference because the Dropout mask introduces additional randomness. The paper defines an inconsistency ratio to measure this difference, with 1 representing perfect consistency. This variance mismatch becomes problematic when the Dropout output is used as input to BatchNorm.

4.

What experiments (architectures, datasets, ablations) demonstrate that the proposed placement outperforms common alternatives, and how large are the gains?

The paper test different Dropout positions on PreResNet-50/110 using CIFAR-10 and CIFAR-100, as well as ResNetV1, Caltech-101, Oxford-IIIT Pet, and ImageNet. P5/P6 generally provide the best results, while earlier positions such as P0–P4 can reduce accuracy substantially. For example, P6 improves CIFAR-100 PreResNet-50 by about 1.0367 points, while the proposed guideline improves ResNetV1-110 on CIFAR-100 by 1.23 points. They also test different weight decay values, bottleneck blocks and several CNN architectures. Overall, the experiments consistently support the proposed placement, although the gains are normal.

5.

What limitations or open questions remain—e.g., does the recommended placement transfer to Transformers, LayerNorm, or non-residual architectures?

The paper mainly focuses on CNN-based residual networks using BatchNorm, so it might not work the same for Transformers or LayerNorm. The theoretical analysis depends specifically on the variance behavior of Dropout and BatchNorm. Therefore, it is unclear whether the same placement rule would work for Transformer architectures, LayerNorm, or completely non-residual networks. The paper also showed that the advantage of placing Dropout before a weight layer depends on certain conditions. Further research is needed to determine whether the proposed guidelines works beyond the normalization methods studied in this paper.